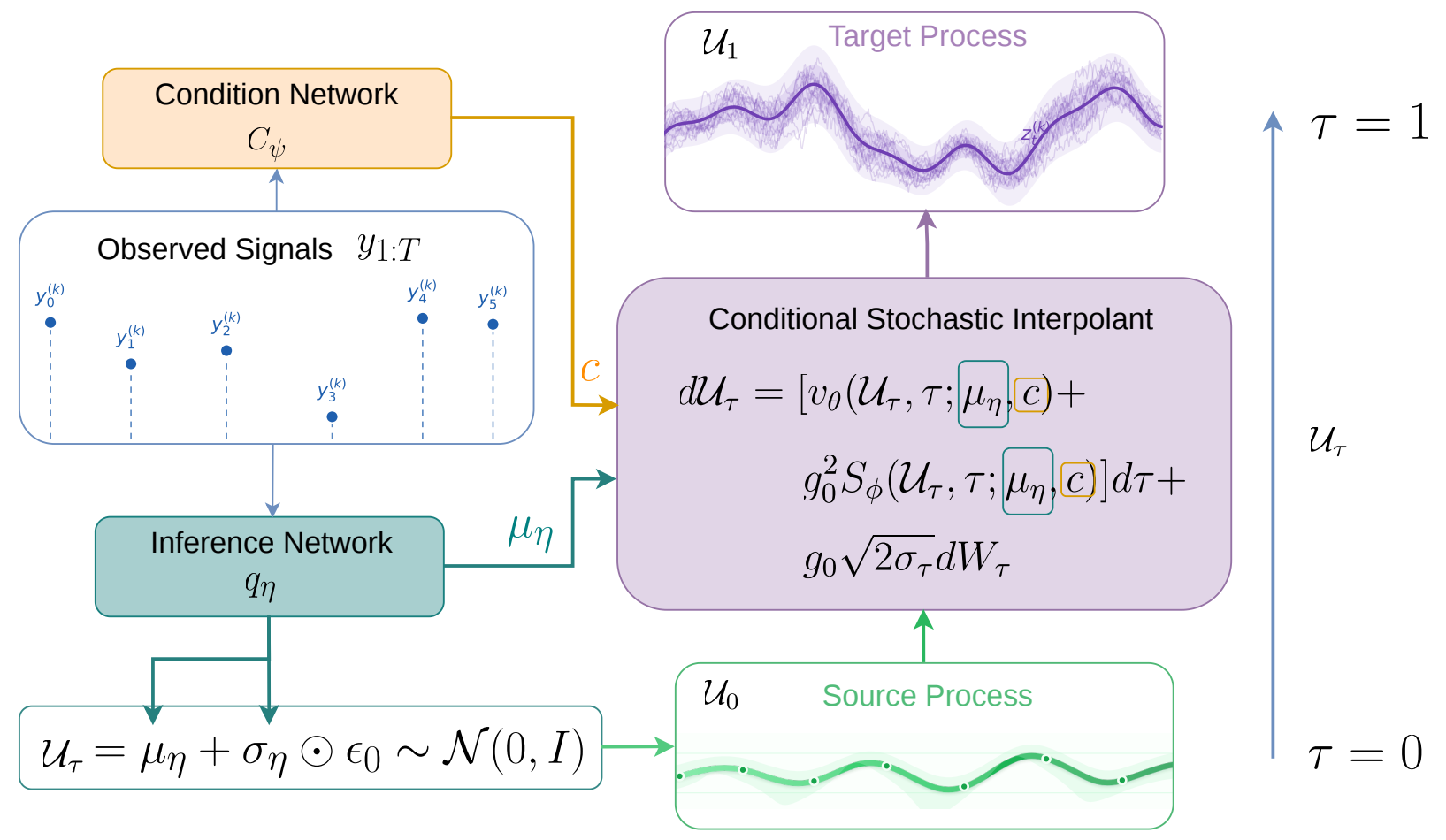

Condition Network
C_ψ
Observed Signals $y_{1:T}$
$y_0^{(k)}$
$y_1^{(k)}$
$y_2^{(k)}$
$y_3^{(k)}$
$y_4^{(k)}$
$y_5^{(k)}$
Inference Network
q_η
$\mathcal{U}_\tau = \mu_\eta + \sigma_\eta \odot \epsilon_0 \sim \mathcal{N}(0, I)$
c
μ_η
$\mathcal{U}_1$
Target Process
$z_t^{(k)}$
Conditional Stochastic Interpolant
$d\mathcal{U}_\tau = [v_\theta(\mathcal{U}_\tau, \tau; \mu_\eta, c) +$
$g_0^2 S_\phi(\mathcal{U}_\tau, \tau; \mu_\eta, c)]d\tau +$
$g_0\sqrt{2\sigma_\tau}dW_\tau$
$\mathcal{U}_0$
Source Process
$\tau = 1$
$\mathcal{U}_\tau$
$\tau = 0$